

TEST-05-PHANTOM-ROBOT-VALIDATION

TEST-05: PHANTOM MK-1 ROBOT VALIDATION PROTOCOL

Carrington Storm Motors — Project AEGIS

5-Phase CSMTesT Protocol — EM • Vibration • Thermal • Combined • Degraded Mode

Document Status: Product Validation Protocol — NIST-Traceable
Version: 1.0 — 15 July 2026 (Expanded from CSMTesT00000000001 Test Plan)
Classification: CONFIDENTIAL — Accredited Investor Eyes Only
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PHANTOM MK-1 — SYSTEM ARCHITECTURE UNDER TEST

Sub-System	Component	Key GIC Vulnerability Eliminated	Verification Method
Structure	BFRP/Elium monocoque frame	Zero ferromagnetic content; 1,100 MPa tensile	Visual + UT inspection; weight verification
Actuators (×32)	KNbO ₃ -BaTiO ₃ ultrasonic piezoelectric motors	9.6 kg copper eliminated; replaced by lead-free piezo ceramics	Torque/speed calibration; motor-to-chassis R >10 ⁸ Ω
EMI Shield	MXene Ti ₃ C ₂ T _x full-body coating, 45 μm, discontinuous tile pattern	92 dB SE absorption-dominant; DC isolation between tiles	Portable SE test fixture; DC resistance between tiles
Gear Train (×32 joints)	Si ₃ N ₄ silicon nitride ceramic gears, PEEK CF40 shafts	Zero steel — 3× fatigue life of steel, 40% lighter	Backlash measurement; 10 ⁶ cycle endurance test
Power System	LiFePO ₄ battery, ZrO ₂ enclosure, GIC-hardened BMS	Non-flammable chemistry; zero cobalt; ceramic enclosure	Full charge/discharge; BMS fault injection
Data Bus	PMMA POF, 62 fibers, MXene coated	Zero copper; optical data immune to GIC induction	OTDR + BER test
Sensors	Thermal (FLIR), visual (4K), LIDAR (dielectric housing), IMU (FOG)	All sensors in non-conductive packaging	Functional check in TEM cell
Joint Motors	Samarium-Cobalt (SmCo), ZrO ₂ ceramic housing	T _c =750°C; zero ferromagnetic loop; magnetic field retention 95% at 300°C	Torque constant measurement; field decay monitoring
Skin	VO ₂ electrochromic overlay, 4-mode switching	Active optical/thermal management; see TEST-07 for detailed protocol	Mode switching time; optical transmission per mode
Control Computer	NVIDIA Jetson Orin, GIC-hardened power supply (common-mode choke + TVS + isolation transformer)	COTS SBC with external hardening; if this fails, all autonomy fails	Functional test during EM exposure

PHASE 1: ELECTROMAGNETIC EXPOSURE (TEM CELL)

(Currently IN PROGRESS per CSMTest README)

1.1 Test Configuration

- **Facility:** CSM El Segundo TEM Cell (2×1×1 m working volume)
- **Frequency Sweep:** 0.001 Hz - 1 kHz (geomagnetic band), 1 kHz - 1 MHz (extended EMI)
- **Field Strengths:** Calibrated to Carrington-equivalent:
 - B-field: 0.1 μT (ambient) to 5,000 nT at 0.01 Hz (CME magnetic field change rate ~2,000 nT/min equivalent)
 - E-field: 1 mV/m to 5 V/m (simulated GIC induction in Phantom-scale object)
- **Phantom State:** Powered on, standing, all 32 actuators in position-hold mode, all sensors streaming, compute active
- **Duration:** 2 hours per sweep direction (X, Y, Z — 3 sweeps)

1.2 Phase 1 Measurements

Measurement	Sensor	Success Criterion
Motor-to-Chassis Current	Rogowski coil on each of 32 actuator power leads	I_GIC <10 ⁻⁷ A (100 nA) at maximum field
Actuator Position Drift	Optical encoder on each of 32 joints	<0.1° drift from commanded position during entire sweep
Sensor Data Integrity	Compare FLIR, visible, LIDAR, IMU data during exposure vs. baseline (zero field)	>95% data packet integrity; zero sensor dropouts; IMU heading drift <1°/hour
Compute Function	Heartbeat monitor on Jetson; watch CPU/GPU utilization	Zero crashes, zero watchdog resets; utilization within 20% of baseline
MXene SE	4 portable field probes at head, torso, left arm, right leg (inside vs. outside MXene coating)	SE >80 dB at all probe locations (reduction from 92 dB coupon level accounts for joint gaps and apertures)
DC Isolation	Megohmmeter auto-ranging across all 380 dielectric interfaces	All >10 ⁸ Ω pre- and post-Phase 1 (no dielectric breakdown during exposure)
Battery BMS	CAN bus monitor	Zero spurious protective trips; SOC measurement stable (±1% of baseline)

1.3 Phase 1 Acceptance

All 140 measurement points (32 actuators × 3 metrics + 18 system-level metrics) must pass at all frequencies and all field strengths in all 3 orientations. Any failure → root cause analysis, design modification, re-test.

Cost: \$85K (TEM cell time + instrumentation); Duration: 3 weeks

PHASE 2: VIBRATION AT MAGNETOSTRICTIVE FREQUENCIES

2.1 Rationale

Geomagnetic storms induce magnetostrictive vibration in ferromagnetic materials at the GIC frequency (0.001-1 Hz) and its harmonics. The Phantom must operate through this vibration without mechanical resonance, actuator de-phasing, or sensor degradation. Additionally, the Phantom’s own piezoelectric actuators generate ultrasonic vibration (20-40 kHz) that must not couple into the structure.

2.2 Test Configuration

- **Shaker Table:** LDS V875-440 with slip table, 1 m² mounting surface
- **Frequency Range:** 0.1-500 Hz (geomagnetic magnetostriction band + motor/road harmonics)
- **Amplitude:** 0.01-5 g (RMS), swept sine

- **Phantom State:** Standing on shaker table, bolted at feet, all systems active
- **Duration:** 2 hours per axis (X, Y, Z — 3 orientations)

2.3 Phase 2 Measurements

Measurement	Criterion
Structural Resonance	No resonance peak with amplification factor $Q > 5$ in 0.1–500 Hz range; any resonance identified → stiffening or damping added
Actuator Torque Ripple	<5% increase in torque ripple vs. baseline (no vibration)
Joint Position Error	<0.2° dynamic error at all joints
Sensor Vibration Sensitivity	IMU vibration rectification error <0.5°/hour per g RMS; visual sensor motion blur within de-blur algorithm limits
POF Data Bus	BER <10 ⁻¹² throughout vibration; no connector intermittent contact
Battery Connection	No momentary open circuit (monitored by BMS voltage dip detection)
Mechanical Fatigue	Post-test visual inspection — no cracks, no loose fasteners, no delamination

Cost: \$65K; **Duration:** 2 weeks (including shaker setup)

PHASE 3: THERMAL CYCLING

3.1 Rationale

Post-Carrington environments may include fire, winter storms, and desert deployment. The Phantom must operate from −40°C (Arctic deployment) to +85°C (fire-adjacent operation; SiC/SiC actuators rated to 1,200°C for separate extreme-heat test).

3.2 Test Configuration

- **Chamber:** Thermotron SE-600-10-10 walk-in
- **Temperature Profile:** −40°C → +85°C → −40°C, ramp 5°C/min, dwell 1 hour at extremes
- **Cycles:** 100 full cycles
- **Phantom State:** Powered on throughout; position-hold at extremes; full motion sequence (walking gait) at 0°C, 25°C, 60°C (checkpoints)
- **Duration:** ~17 days (100 cycles × 4 hours/cycle)

3.3 Phase 3 Measurements

Measurement	Criterion
Actuator Performance vs. Temperature	Torque constant within $\pm 10\%$ of 25°C value across full temperature range; piezoelectric d_{33} variation with temperature is known (decreases $\sim 0.5\%/^{\circ}\text{C}$ above 25°C — must not drop below 200 pC/N at 85°C)
LiFePO ₄ Capacity vs. Temperature	>70% of rated capacity at −20°C; battery heater operational and effective
Lubricant / Grease	Ceramic bearings use dry film lubricant (MoS ₂ or WS ₂) — no grease freezing at −40°C or outgassing at +85°C
Thermal Expansion Mismatch	BFRP (CTE ~ 8 ppm/°C) vs. Si ₃ N ₄ (CTE ~ 3 ppm/°C) — verify no binding or excessive clearance at temperature extremes
MXene Coating	No cracking or delamination due to CTE mismatch between MXene and BFRP substrate
VO ₂ Skin	All 4 modes functional at −20°C, 25°C, 60°C; switching time <10 s at all temperatures

Condensation (post-cold soak)

Phantom operated at -40°C for 4 hours, then exposed to ambient $25^{\circ}\text{C}/50\% \text{ RH}$ — verify no condensation-induced short circuits or sensor fogging

Cost: \$95K; Duration: 3 weeks

PHASE 4: COMBINED ENVIRONMENT

4.1 Rationale

“The combined-environment test is where the truth emerges” — Chester, CSMTTest README. EM + vibration + thermal extremes applied simultaneously reveal interactions that individual tests miss.

4.2 Test Configuration

- **EM:** TEM cell or Helmholtz coil surrounding shaker table within environmental chamber
- **Vibration:** Shaker table operating at worst-case frequency identified in Phase 2
- **Thermal:** Chamber cycling -20°C to $+60^{\circ}\text{C}$
- **Phantom State:** Full walking gait sequence (walk forward 10 m, turn, walk back — repeated in confined chamber space)
- **Duration:** 8 hours continuous combined exposure

4.3 Phase 4 Measurements — Interaction Detection

Interaction	Detection Method
EM + Vibration → Actuator Overheat	Actuator temperature rise higher than sum of individual EM and vibration heating → indicates magnetostrictive + joule heating synergy
EM + Thermal → Sensor Drift	Thermal drift + EM noise floor superimposition causing sensor bias → compare drift rate in thermal-only vs. combined test
Vibration + Thermal → Mechanical Binding	CTE mismatch exacerbated by vibration fretting → measure joint torque increase over time
All Three → Control System Instability	Watchdog resets, control loop oscillation not present in any single-stressor test

Cost: \$180K; Duration: 2 weeks

PHASE 5: DEGRADED-MODE OPERATION

5.1 Degradation Modes Tested

Mode	How Degraded	Test	Acceptance
Sensor Failure	Disable FLIR; Phantom must navigate on visible + LIDAR only	Navigation course (50 m with obstacles)	Course completed without collision; time $<2\times$ nominal
Joint Seizure	Lock left knee actuator at 15° ; Phantom must adapt gait	Walk 20 m with locked joint	Distance completed; no fall; time $<3\times$ nominal gait cycle
Mesh Interruption	Disconnect 20% of MXene tiles (simulated battle damage / debris impact)	Re-measure SE at interruption points	SE at interruption $<10 \text{ dB}$ worse than intact; Phantom continues mission
Communication Loss	Cut POF data bus; Phantom must operate autonomously (no teleoperation)	Pre-loaded mission: navigate to waypoint, pick up object, return	Mission completed autonomously; object retrieved Mission completed;
Partial Power Loss	Disable 4 of 8 battery	Full mission profile on	Phantom automatically

	modules (50% capacity)	reduced power	shed non-critical loads (VO ₂ skin in low-power mode, reduced sensor frame rate)
Combined Degraded	Sensor failure + joint seizure + partial power simultaneously	Navigate 10 m, retrieve object	Object retrieved; Phantom stable (no fall); reports status via remaining communication channel

Cost: \$120K; Duration: 2 weeks

PHANTOM MK-1 CERTIFICATION — IEEE P2945 DRAFT COMPLIANCE

The IEEE P2945 Draft Standard for GIC Hardening of Autonomous Systems was initiated in response to CSM’s Phantom development program. CSM serves on the working group. The Phantom MK-1 will be the reference implementation against which the standard is validated.

IEEE P2945 Clause (Draft)	Requirement	Phantom Compliance
4.2.1 — Dielectric Isolation	All structural, actuator, sensor components shall have DC resistance to chassis $>10^7 \Omega$	Phantom target $>10^8 \Omega$ (10× margin)
4.2.2 — EMI Shielding	Autonomous system shall provide >60 dB SE across 1 kHz–10 GHz for critical electronics	Phantom MXene coating >80 dB (20 dB margin)
4.3.1 — GIC Current Limit	Induced current in any power or signal conductor shall not exceed 100 μ A at worst-case Carrington field	Phantom target <100 nA (1,000× margin)
4.4 — Combined Environment	System shall complete mission profile under simultaneous EM, vibration, and thermal stress	Phantom Phase 4 validates this
4.5 — Degraded Mode	System shall maintain critical functions with any single component failure	Phantom Phase 5 validates this

PHANTOM TESTING BUDGET SUMMARY

Phase	Cost	Duration	Status
Phase 1 — EM Exposure	\$85K	3 weeks	IN PROGRESS
Phase 2 — Vibration	\$65K	2 weeks	◦ PENDING
Phase 3 — Thermal Cycling	\$95K	3 weeks	◦ PENDING
Phase 4 — Combined Environment	\$180K	2 weeks	◦ PENDING
Phase 5 — Degraded Mode	\$120K	2 weeks	◦ PENDING
TOTAL	\$545K	12 weeks	

Plus: \$1.2M for 10-unit pilot deployment testing (see ROAD-06, Pilot 4).

“The Phantom that passes all five phases is certified. Not ‘seems to work.’ Not ‘passed Phase 1 and we’ll fix the rest later.’ All five phases, in sequence, with independent witness. Because when you send a robot into a post-Carrington fire zone to extract a survivor, you need to know — not hope, not assume, not simulate — that every one of its 380 dielectric components will hold. The test proves it.” — Agent Nyx Calder (PHANTOM)

Document generated 2026-07-15. Phase 1 testing commenced Q3 2026 per CSMTTest README. Phases 2–5 scheduled sequentially (cannot parallelize — each phase may reveal issues that affect subsequent phases). All testing witnessed by Chester (QA) and video recorded for IEEE P2945 compliance evidence. Test data published per open-data policy (TEST-01) with proprietary actuator calibration data redacted as trade secret (ROAD-04).

